Internship Report: ShadowFox Virtual Internship

Role: Data Analyst | Level: Advanced | Duration: 1 September to 30 September 2026

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1. Introduction

I joined the ShadowFox Virtual Internship as a Data Analyst. I worked through three levels. At the beginner level I built an Excel sales dashboard. At the intermediate level I analysed customers and revenue. At the advanced level I built an executive-style HR attrition dashboard in Power BI.

2. Beginner and Intermediate Summary

- Beginner: cleaned a retail sales dataset and built KPIs, trend charts and a dashboard in Excel.
- Intermediate: studied repeat customers, revenue concentration, segments and seasonality, and gave recommendations.

3. Advanced Level: Objectives

- Clean and transform an HR dataset in Power Query.
- Choose KPIs and plan the dashboard layout.
- Build interactive pages with slicers, drill-down and drill-through.
- Tell a business story with insights and recommendations.

4. Dataset and Tools

I used an HR analytics sample dataset with the IBM HR Analytics columns: 1470 employees and 35 columns. It is practice data, not real company data. I used Power BI Desktop, Power Query and DAX. The work is saved on GitHub.

5. Data Cleaning and Transformation

- Checked nulls and duplicates in Power Query. None were found.
- Removed the constant columns EmployeeCount, Over18 and StandardHours.
- Added AttritionFlag, AgeBand, IncomeBand and TenureBand with DAX.
- Built 13 measures, including Attrition Rate, Overtime Attrition Rate and Gap vs Overall.

6. Dashboard Design

The dashboard has three pages. The Executive Overview page has six KPI cards, five slicers (Department, Gender, Job Role, Age Band, Overtime) and charts by department, age band and job role. The Attrition Drivers page compares attrition by overtime, travel, income, tenure and job satisfaction. The Employee Details page is a drill-through table. A Department to Job Role hierarchy supports drill-down.

7. Key Findings

Area	Result
Headcount / Leavers	1,470 / 238
Attrition rate	16.2%
Overtime effect	26.6% with overtime vs 12.2% without
Department	Sales 20.4%, R&D 14.9%, HR 7.0%
Age	Under 30: 27.4%. Age 50+: 10.0%
Income	Under 3K: 25.3%. 10K+: 4.2%
Job role	Sales Representative 32.2%. Manager 3.4%
Travel	Frequent 23.6%. Rare 14.9%
Pay of leavers vs stayers	USD 4,850 vs USD 7,500 per month

8. Recommendations

- Limit overtime and add staff in teams with heavy overtime.
- Review pay in the lowest income bands.
- Give early-career and Sales staff clear growth paths.
- Reduce frequent travel where possible.
- Track attrition monthly with the dashboard.

9. Challenges and Learning

The main challenge was choosing KPIs that help a manager decide quickly. I learned to write DAX measures, to build bands for age, income and tenure, and to use slicers and drill-through. I also learned to turn charts into clear recommendations.

10. Conclusion

This internship gave me practice in the full analytics flow, from raw data to an executive dashboard. I plan to use these skills in my academic projects.

GitHub repository: [paste your repository link here]